

Amazon EMR – Complete Overview

What is Amazon EMR?

Amazon EMR (Elastic MapReduce) is a **cloud-native big data platform** that helps process and analyze massive volumes of data using **open-source tools** like:

- Apache Spark
- Apache Hadoop
- Hive, HBase, Presto, Flink, and more

 **Primary Goal:** Run **distributed data processing frameworks** quickly and efficiently at scale.

Key Concepts

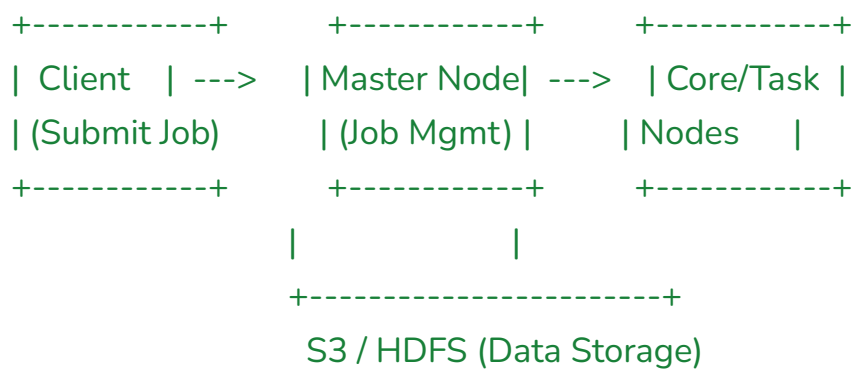
Concept	Description
 Distributed Processing	Uses a cluster of EC2 instances to process large data workloads in parallel
 Open-Source Frameworks	Supports Spark, Hadoop, Hive, HBase, Presto, and others
 Managed Service	AWS handles provisioning, configuration, and scaling of resources

-  **Decoupled Storage** Works with Amazon S3 instead of relying on HDFS
 -  **Pay-As-You-Go** You pay for compute and storage resources per second
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Architecture

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◆ Node Types

Node Type	Role
 Master Node	Manages cluster, runs job tracker, monitors health
 Core Node	Runs task and stores data (HDFS or S3)
 Task Node	Only runs task, no data stored (stateless)

Supported Frameworks

Tool	Purpose
 PySpark	Python-based distributed data processing framework
 Docker	Containerization technology for running applications in isolated environments
 Kubernetes	Container orchestration system for automating deployment and scaling of containerized applications

 Apache Spark	Fast in-memory analytics
 Hadoop MapReduce	Traditional batch processing
 Apache Hive	SQL-like queries on big data
 HBase	NoSQL database on HDFS
 Presto	Fast interactive SQL on large data
 Flink	Real-time streaming analytics

Data Sources & Storage

- **Amazon S3** – Primary storage (most cost-efficient, scalable)
 - **HDFS** – Hadoop-native file system (optional)
 - **DynamoDB, RDS** – External data sources
 - **Glue Catalog** – Metadata integration
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Key Features

Feature	Description
 Autoscaling	Automatically scale nodes up/down



EMR Notebooks

Jupyter-based interface for Spark jobs



Security

IAM, KMS encryption, VPC, Kerberos, SSL



Monitoring

CloudWatch + EMR Console



Bootstrap Actions

Run custom scripts at instance launch



Step Execution

Sequential tasks like Spark job →
Hive query



Use Cases

-  Data transformation & ETL jobs
-  Machine learning preprocessing (e.g., with Spark MLlib)
-  Log analysis (Apache logs, clickstream, etc.)
-  Data science and experimentation
-  E-commerce behavioral analysis
-  Social media or sensor data processing